



VASANTDADA PATIL PRATISHTHAN'S
COLLEGE OF ENGINEERING AND VISUAL ARTS
An Autonomous Institute | Affiliated to University of Mumbai
NAAC 'A' Grade Accredited | ISO 9001:2015 Certified



Department of Information Technology
NBA Accredited (Dated 01/07/2024 to 30/06/2027)

VIVA CLEARED (3/3)

STUDENT **KONKA LOKESH SHANKAR LAVANYA**

SUBJECT

**Exp 1: Python and ML Environment set up and
Data Exploration and visualization**

ENROLLMENT NO **vu4f2425008**

DATE **18/08/2026**

theory

Python is one of the most popular programming languages used in Machine Learning (ML), Data

Science, Artificial Intelligence (AI), Data Analytics, and Scientific Computing. It provides a large number of libraries that simplify mathematical calculations, data manipulation, visualization, and machine learning model development.

A proper Python environment is necessary before performing any ML task. Google Colab provides a cloud-based environment where Python programs can be executed without installing software on a local computer.

What is Jupyter Notebook?

Jupyter Notebook is an open-source web-based interactive programming environment where

code, text, equations, graphs, and outputs can be written in a single notebook.

Features

- Interactive coding
- Easy debugging
- Rich visualization support
- Supports Markdown documentation
- Suitable for Data Science and ML
- Executes code cell by cell

Advantages

- Beginner-friendly
- Easy visualization
- Supports multiple programming languages

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- Saves notebooks (.ipynb)
- Excellent for experiments and reports

Essential Python Libraries

1. NumPy

NumPy (Numerical Python) is used for numerical computations and multidimensional arrays.

Features

- Fast mathematical operations
- Matrix calculations
- Statistical functions
- Linear algebra
- Random number generation

Example

```
import numpy as np
a=np.array([10,20,30])
print(a)
```

2. Pandas

Pandas is used for data manipulation and analysis.

Features

- DataFrame
- Series
- Reading CSV files
- Cleaning data
- Filtering
- Grouping
- Sorting

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Example :

```
import pandas as pd
```

3. Matplotlib

Matplotlib is the basic plotting library.

Used for

- Line Graph
- Bar Chart
- Histogram
- Scatter Plot
- Pie Chart
- Box Plot

Example :

```
import matplotlib.pyplot as plt
```

4. Seaborn

Seaborn is built on Matplotlib and provides attractive statistical graphs.

Features

- Heatmaps
- Pairplots
- Countplots
- Distribution plots
- Regression plots
- Better themes

• Better themes

Example

```
import seaborn as sns
```

Reproducible Development

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Reproducibility means obtaining the same output every time an experiment is executed.

It can be achieved by:

- Using fixed random seed
- Saving datasets
- Using same library versions
- Writing clean code
- Documenting every step

Example

```
np.random.seed(100)
```

algorithm

Logic: https://drive.google.com/file/d/19x2gRQPWfP0dAd-4rUfRgKyj3KCqkyIp/view?usp=drive_link

conclusion

Hence, we successfully executed the Python and Machine Learning (ML) environment setup by establishing a Python environment using Jupyter Notebook/VS Code and installing the essential ML libraries for reproducible development.

output

Output: https://drive.google.com/file/d/19x2gRQPWfP0dAd-4rUfRgKyj3KCqkyIp/view?usp=drive_link

ASSESSMENT SUMMARY

Criteria	Max Marks	Obtained
Attendance	3	0
Discipline	3	0
Short Oral	3	0
Correctness of Lab Report	3	0
Timely Completion of Lab Report	3	0
TOTAL	15	0



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